Jaya Agnihotri

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Objective

I aim to contribute to the advancement of numerical methods while utilizing my skills in mathematical modeling and high-performance computing. My research focuses on designing positivity-preserving and divergence-constraint-preserving finite difference schemes for two-fluid plasma flows.

Education

Postdoc	Institute of Mathematics, TU-Clausthal, Germany • **Advisor**: **Prof. Dr. Philipp Öffner**	Sept 2025 - Till Date
PhD, Mathematics	Indian Institute of Technology Delhi, Delhi, India • Numerical Analysis, Hyperbolic Conservation laws, Scientific Computing. • **Advisor**: **Prof. Harish Kumar**	2019 – 2025
Master of Science	Indian Institute of Technology (Indian School of Mines), Dhanbad, India	2017 - 2019
Bachelor of Science	CSJM University, Kanpur, India	2012 - 2015

Publications and Preprints

- *Second order divergence constraint preserving entropy stable finite difference schemes for ideal two-fluid plasma flow equations.* **Journal of Scientific Computing**, 101**(2024)**, Article no. 46, doi: https://doi.org/10.1007/s10915-024-02685-0.
- *Second order divergence constraint preserving entropy stable finite difference schemes for two-fluid relativistics plasma flow equations.* (**Communications on Applied Mathematics and Computation**).
- *Robust divergence constraint preserving second-order schemes for two-fluid plasma flow equations.* (**To be communicated**).
- *Convergence analysis of multicomponent flows.* (**In Progress**).

Talks

- Contributed talk in the **14th AIMS conference** held at Abu Dhabi, UAE, December 16 - 20, 2024 AIMS2024.
- Contributed talk in the 19th international conference entitled "Hyperbolic Problems: Theory, Numerics and Applications (**HYP 2024**)" held at Shanghai Jiao Tong University, PR China, July 1 - 5, 2024. HYP2024
- Contributed talk in the international conference entitled "International Conference on Applied Mathematics and Mechanics (**ICAMM 2023**)" held at IIT Indore, India, October 18 - 20, 2023 ICAMM2023.
- Contributed talk in the 38th Annual Conference of Ramanujan Mathematical Society, (**RMS 2023**) held at IIT Guwahati, India, December 22 - 24,2023 RMS2023.
- "**NCMW** Finite volume and spectral methods for hyperbolic problems" held at TIFR-CAM Banglore, December 4 - 15, 2023 NCMW2023.
- Webinar series on broad area of Partial Differential Equations (PDEs) held by IIT-K in joint collaboration with TIFR-CAM and IISER-PUNE, IISER-KOLKATA, (Dec 2020).
- NSM Workshop on High Performance Computing (HPC) for Computational Fluid Dynamics (CFD), Organised by Indian Institute of Technology Bombay and Centre for Development of Advanced Computing (C-DAC) (May 2022).

Teaching Experience

Numerical Methods and Computations (MTL 107), Teaching Assistantship, IIT Delhi.	Fall, 2024
Numerical Methods and Computations (MTL 107), Teaching Assistantship, IIT Delhi.	Spring, 2024
Calculus (MTL 100), Teaching Assistantship, IIT Delhi.	Fall, 2023
Differential Equations (MTL 102), Teaching Assistantship, IIT Delhi.	Spring, 2023
Numerical Methods and Computations (MTL 107), Teaching Assistantship, IIT Delhi.	Fall, 2022
Calculus (MTL 100), Teaching Assistantship, IIT Delhi.	Spring, 2022
Numerical Methods and Computations (MTL 107), Teaching Assistantship, IIT Delhi.	Fall, 2021
Linear Algebra and Differential Equations (MTL 101), Teaching Assistantship, IIT Delhi.	Spring, 2021

Computer Skills

- **Languages**: C, C++ and Python.
- **Mathematical**: Mathematica, MATLAB.
- **CPU Parallel Computing**: PetSc, OPEN-MPI, Shared Memory(OPENMP).
- **IDLs**: Vim, VS-Code.

Personal Details

Language:	English, Hindi, German(Beginner)
Gender:	Female
Address:	TU-Clausthal, Institute of Mathematics Numerical Analysis Group Room 206 / Erzstraße 1 38678 Clausthal-Zellerfeld, Germany